

$$F = C - P + 2$$

$$= 1 - 1 + 2$$



THE PHASE RULE

"It is difficult to say what is impossible, for the dreams of yesterday is the hope of today and the reality of tomorrow"

Robert Goddard

CHAPTER OUTLINE

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|-----|---|-----|---|
| 2.1 | Definitions | 2.5 | Two Component System |
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| 2.3 | Phase Diagram | 2.7 | Classification of two Component Solid-liquid Equilibria |
| 2.4 | One Component System | 2.8 | Simple Eutectic System two Component System which from a General Phase Diagram |
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Water $\left\{ \begin{array}{l} \text{Solid} \\ \text{Gas} \end{array} \right. \rightarrow \text{all are phases} \rightarrow \text{change from one phase to another is phase transition}$

A substance can exist in various states depending on the external conditions. For example, water can exist in the form of a solid liquid and vapour. Each of the state is referred to as a phase. A change in the state of substance is called a phase transition. Phase transitions occur when energy (usually in the form of heat) is added or removed from a particular phase. A phase transition, like any other thermodynamic system, proceeds until equilibrium is established. Before discussing the details, let us explain the various terms we would be using in the course of this chapter.

2.1 DEFINITIONS

Phase

A phase (P) is a homogeneous physically distinct and mechanically separable part of a system which is separated from other parts of the system by definite boundaries.

In other words, phase is a state of matter that is uniform throughout, not only in chemical composition but also in physical state.

A system consisting of one phase only is called a homogeneous system and a system consisting two or more phases is called a heterogeneous system.

Phase

Homo Hetero
1 phase 2/ More than 2.

(i) A crystal of a substance is a single phase. But if the crystals of two substances are mixed then there exist two phases.

Ans (ii) A gas or a gaseous mixture consist of a single phase, since there is no demarcation (interface) between one gas and another. Air is in one phase only although it is a mixture of number of gases.

(iii) A solution of two miscible liquids will exist in one phase only as far as solution is concerned. But the system as a whole will have two phases - one phase will be the solution (liquid) and the other would be the vapour above it.

(iv) If two immiscible liquids are mixed together to form a solution, then the total phases present would be three. Two phases are for the substance in liquid and the third one for vapours, containing the vapours of both the liquids.

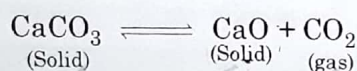
(v) Solution of sodium chloride in water is a uniform solution. Therefore, it is a one phase system.

(vi) Saturated solution of water and sugar has two phases (solution of water + sugar and solid sugar settled at the bottom).

(vii) Sulphur exists in two allotropes in nature. As these allotropes are similar in chemical properties but are different in physical properties, therefore a mixture of two allotropes is a two phase system.

(viii) A mixture of two or more chemical substances contains as many phases. Each substance has different physical and chemical properties. Therefore a mixture of CaCO_3 and CaO is a two phase system.

(ix) When CaCO_3 is heated in a closed vessel, it decomposes to CaO and CO_2 . At equilibrium,



At this state there are two solid and one gaseous phase. Therefore, it is a three phase system.

Components

Minimum no. of independent constituents required to define a system is called component.

The number of components (C) of a system is the smallest number of independently variable constituents at equilibrium, by means of which the composition of each phase present can be expressed either directly or in the form of a chemical equation. Zero and negative quantities of the components are acceptable when composition of a phase is expressed in terms of the components.

The components of a system do not express the number of constituents or chemical entities or chemical individuals present in the system.

To understand the meaning of component and to find the number of components of a system following points are noteworthy.

(i) The chemical formula representing the composition of a phase is written on left hand side (LHS).

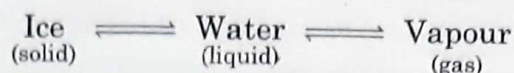
(ii) The remaining chemical constituents existing independently in the system as represented by chemical formulae are written on right hand side (RHS).

(iii) The quantities of constituents on RHS can be made minus (-) or zero (0) to get the composition of the phase on LHS.

All liquids has some vapour pressure. Vapour press. is a strong function of the temperature. So some liq. have significant vapor pressure at room temp while don't. (are volatile).

Examples

- (i) Consider a water system. At equilibrium there are three phase



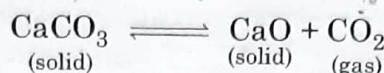
It is a one component system because each phase can be expressed in terms of only one component, i.e. water (H_2O).

- (ii) In sulphur system there are four phases namely rhombic sulphur, monoclinic sulphur, liquid and vapour.



It is a one component system as the composition of each phase can be expressed in terms of sulphur only.

- (iii) Decomposition of CaCO_3 : = 2 component system.



The system consists of three phases – solid CaCO_3 , solid CaO and gaseous CO_2 . Due to existence of equilibrium, composition of each of the three phases can be expressed in terms of any two constituents, CaCO_3 , CaO and CO_2 as shown below:
If CaCO_3 and CaO constituents

Phase	Component
CaCO_3 (s)	$\text{CaCO}_3 + 0\text{CaO}$
CaO (s)	$0\text{CaCO}_3 + \text{CaO}$
CO_2 (g)	$\text{CaCO}_3 - \text{CaO}$

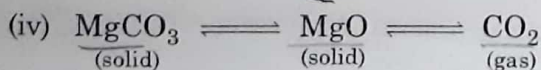
If CaCO_3 and CO_2 constituents are chosen

Phase	Component
CaCO_3 (s)	$\text{CaCO}_3 + 0\text{CO}_2$
CaO (s)	$\text{CaCO}_3 - \text{CO}_2$
CO_2 (g)	$0\text{CaCO}_3 + \text{CO}_2$

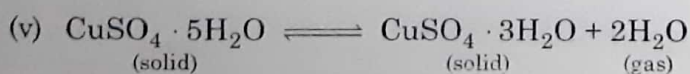
If CaO and CO_2 constituents are chosen

Phase	Component
CaCO_3 (s)	$\text{CaO} + \text{CO}_2$
CaO (s)	$\text{CaO} + 0\text{CO}_2$
CO_2 (g)	$0\text{CaO} + \text{CO}_2$

Thus in all the cases, only two components are required to express the composition of each phase. Hence, it is a two component system and is also known as binary system.

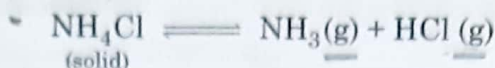


It is also a binary system or a two component system because to specify the composition of any phase, we need only two constituents.



It is also a binary system.

(vi) In the dissociation of ammonium chloride in vacuum, equilibrium occurs as follows.



The system consists of two phases, solid NH_4Cl and a gaseous mixture of NH_3 , HCl and NH_4Cl . The composition of either of the two phases can be expressed in terms of one component as NH_3 and HCl are formed in fixed stoichiometric proportions by the reaction. Therefore both the phases have formal composition ' NH_4Cl '. However, if NH_3 or HCl is in excess, the system becomes a two component system because decomposition of NH_4Cl would not give the correct composition of the gas phase. Hence NH_3 or HCl would be required to specify the gas composition. Hence it becomes two component system. If there are ' S ' substances and ' R ' relations (equations representing chemical equilibria) among them, then the number of components (C) can be calculated as:

$$C = S - R$$

For example, consider a solution of glucose in water. There are two substances and hence $S = 2$, but there is no relation between them so $R = 0$, therefore.

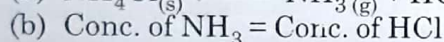
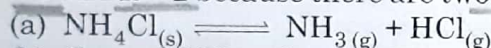
$$C = S - R$$

or

$$C = 2 - 0 = 2$$

Hence, it is a two component system.

Now consider $\text{NH}_4\text{Cl}_{(s)}$ in equilibrium with its vapours containing NH_3 and HCl . Here $S = 3$ and $R = 2$ because there are two relations as shown,



Therefore $C = S - R = 3 - 2 = 1$. Thus it is one component system.

(vii) Gaseous mixture O_2 and N_2 is a two component system because in the gaseous mixture O_2 and N_2 are two different chemical entities to express the mixture.

Phase	Component
Gaseous mixture of O_2 and N_2	$x \text{O}_2 + y \text{N}_2$

(viii) The aqueous solution of NaCl constitutes one phase only but to determine the composition of this homogeneous phase ($x \text{NaCl}$ $y \text{H}_2\text{O}$) two chemical entities are required. Therefore it is a two component system.

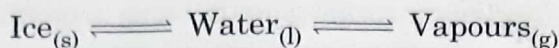
Phase	Component
Aqueous solution of NaCl	$x \text{NaCl} + y \text{H}_2\text{O}$

DEGREE OF FREEDOM

The degree of freedom (F) of a system is defined as the number of independent variables, such as temperature, pressure and composition which are to be specified to describe a system completely. It is also termed as variance.*

*A system having 1, 2, 3 or 0 degrees of freedom is called univariant, bivariant, trivariant and non variant or invariant respectively.

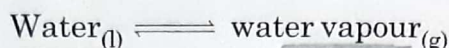
- (i) For a system consisting of Ice, water and vapours,



phase = 3
component = 1

if all the three phases are present in equilibrium, then no condition has to be specified. This is because the three phases of water can exist together in equilibrium only at a particular temperature and pressure (corresponding to the freezing point) and no condition is necessary to be specified to define the system. Thus it is a non-variant system or invariant system, i.e. $F = 0$.

- (ii) For a system consisting of water in contact with its vapour, i.e.



Such a system can be completely defined by specifying either pressure or temperature. This is because on fixing one variable, the other becomes automatically fixed. Hence, either temperature or pressure is required to specify the system. The system is therefore, univariant i.e. $F = 1$.

- (iii) For a system consisting of water vapours only, both temperature and pressure are to be specified in order to describe the system completely. Thus it is a bivariant system i.e. $F = 2$.
- (iv) For a pure gas, $F = 2$. This is because $PV = RT$, if the value of pressure (P) and temperature (T) are fixed, volume automatically gets fixed.
- (v) A system of saturated NaCl solution has $F = 1$. Because if only one variable, i.e. temperature is fixed, composition and vapour pressure get automatically fixed.

2.2 PHASE RULE

The phase rule is an important generalisation and a versatile tool which is related to the study of behaviour of heterogeneous systems. In general, it may be stated that with the help of phase rule, it has been possible to predict quantitatively the effect of changing pressure, temperature and concentration on a heterogeneous system in equilibrium by means of a phase diagram.

American physicist Willard Gibbs (1874) deduced this generalisation on the basis of thermodynamics and H.W.B. Roozeboom (1884) explained it further.

Gibbs Phase Rule

Gibbs phase rule states that if a heterogeneous system is influenced by temperature, pressure and concentration and not by any other action like gravity, electrical, magnetic forces or by surface action, then the sum of number of phases (P) and degrees of freedom (F) is greater than the number of components (C) by two. It is expressed mathematically as follows:

$$\boxed{P + F = C + 2}$$

or

$$\boxed{F = C - P + 2}$$

$$F = 1 - 3 + 2$$

where P is the number of phases.

C is the number of components and

F is the number of degrees of freedom

Derivation of the Phase Rule

Consider a heterogeneous system in equilibrium consisting of 'C' components distributed in 'P' phases. We have to determine the degrees of freedom of this system.

The number of degrees of freedom (F) of a system in equilibrium is the number of variables (such as temperature, pressure and composition) that must be arbitrarily fixed to define the system completely. Now, let us find out how many of number of variables are there in the above system.

- When the system is in equilibrium, there can be only one temperature and one pressure, hence the total number of these variables is = 2
- The number of composition variables can be more. For each phase the number of composition variable will be $(C - 1)$ because the composition of remaining component can be found out by difference (i.e. for example, out of 10 components required to describe the composition of a phase, if the mole fraction of 9 are known, remaining one can be found out by the difference)

Now, the number of phases = P

Number of composition variable in each phase = $(C - 1)$

Total number of composition variables in all phases = $P(C - 1)$

Number of other (temperature and pressure) variables = 2

Total number of variables of the system = $P(C - 1) + 2$

- Now we have to find the total number of relations among the variables, i.e. number of equations possible at equilibrium.

When a heterogeneous system is in equilibrium at constant temperature and pressure, the chemical potential of a particular component must be same in all phases in which it appears. For a one component system, having two phases α and β , the equality of chemical potential implies

$$\mu(\alpha) = \mu(\beta)$$

where $\mu(\alpha)$ and $\mu(\beta)$ are the chemical potentials of the two phases α and β respectively. Similarly, if there is one component in three phases (say α , β and γ) and one of these phases is referred to as standard phase, then can be expressed in the form of following two equations

$$\mu_\alpha = \mu_\beta$$

$$\mu_\alpha = \mu_\gamma$$

Thus, for each component in equilibrium in three phases, two equations are possible.

Hence, for each component in 'P' phases $(P - 1)$ equations can be written.

Since, there are 'C' components, the number of equations or variables possible at equilibrium are:

$$C(P - 1)$$

Therefore the degrees of freedom or number of independent variables F is given by

$$F = (\text{Total number of variables} - \text{Total number of relations among the variables})$$

$$= [P(C - 1) + 2] - [C(P - 1)]$$

$$\therefore F = C - P + 2$$

Advantages of Phase Rule

- It provides a simple method of classifying equilibrium states of systems such as phases components and degrees of freedom.
- It indicates that different systems having the same number of degrees of freedom behave in like manner.
- It explains the behaviour of systems when allowed to changes in the variables such as pressure temperature and concentration.
- The phase rule is applicable to macroscopic systems. Therefore it is not compulsory to take in to account their molecular structures.

- (v) It is applicable to physical as well as chemical equilibria.
- (vi) Phase rule predicts that under a given set of conditions.
 - Whether various substances would exist together in equilibrium. ✓
 - Whether some of the substances would be interconverted ✓
 - Or whether some of the substances will completely disappear.

Limitations of Phase Rule

- (i) As the phase rule is applicable to heterogeneous systems in equilibrium, it is therefore of no use for such systems which are slow in reaching the state of equilibrium.
- (ii) As the phase rule is applicable to a single equilibrium state, it never tells about the number of other equilibrium possible in the system.
- (iii) In phase rule the different variables are pressure, temperature and concentration or composition. But phase rule does not consider the other factors like electric and magnetic influences.
- (iv) All the phases are required to be present under the same temperature, pressure and gravitational force.
- (v) Solid or liquid phases should not be in finely divided form, otherwise vapour pressure may differ from the normal values.
- (vi) It does not take into consideration the quality of the phases though it takes into account the number of phases.

Application of Phase Rule

All the systems are classified on the basis of number of components present. Thus we may have one, two, three, etc. component systems. The phase rule is perfectly applicable to these systems. The phase diagram is an important medium to clearly indicate the equilibrium conditions between different phases in a system. The phase diagram is helpful for studying and controlling various processes such as phase separation, solidification of metals, the change of structure during heat treatment, etc.

2.3 PHASE DIAGRAM

A phase diagram is a graphical representation of the pressure and temperature under which two or more physical states can exist together in a state of dynamic equilibrium. Phase diagram illustrates the conditions of equilibrium between various phases of a substance. If the temperature is plotted against pressure the diagram is called P-T diagram. The study of phase diagram is vital for proper understanding of the relative stability of equilibrium between phases. With the help of

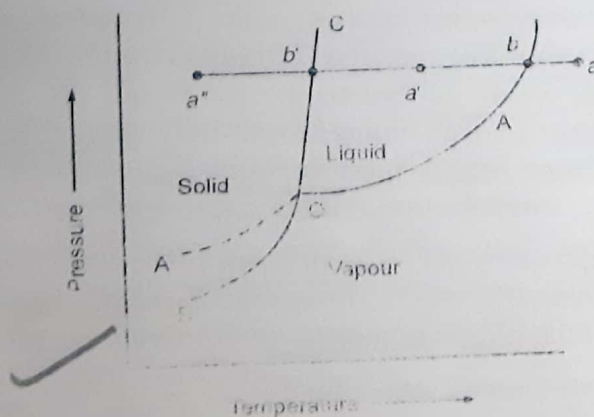


Fig. 2.1 A typical phase diagram of a one component system

knowledge of the number of phases and components of the system, phase rule enables one to predict clearly the conditions under which a system can remain in equilibrium.

Fig. 2.1 is a typical case for a *one-component systems*. The diagram consists of:

- (a) Areas (b) Curves/Lines
(c) Triple point

- (a) **Areas:** The diagram has 3 regions or areas labelled as solid, liquid and vapour, i.e. COB, COA and BOA respectively. Each of the three areas shows the conditions of temperature and pressure under which the respective phase can exist. When only one phase is present,

$$F = 1 - 1 + 2 = 2$$

Thus, each phase has two degrees of freedom; both temperature and pressure must be specified to define the state of the system. Consider a point *a* in the Fig. 2.1. At this point the gas phase of the substance is thermodynamically the most stable state. If the temperature is reduced at constant pressure, the system moves to point *b* where the liquid and its vapour are in equilibrium. If the temperature is still reduced further it will reach a point *a'* where liquid is the stable phase. On reducing the temperature further point *b'* is reached, where liquid and solid phase are in equilibrium. On further reducing the temperature upto *a''*, solid is the stable phase.

- (b) **Curves:** There are three lines or curves separating the areas. These show the conditions of equilibrium between any two of the three phases:

These are also known as phase boundaries.

- Curve OA represents the equilibrium,



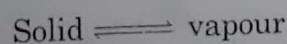
This is referred to as **vapour pressure curve**. Above the curve liquid is stable phase while below it, vapour is the stable phase.

- Curve OC represents the equilibrium



This is referred to as **melting curve** or **fusion curve** or **freezing point curve** and indicates the influence of pressure on melting point of solid.

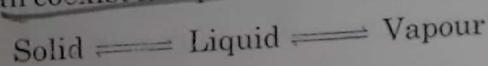
- Curve OB represents the equilibrium



This is referred to as **sublimation curve**.

- **Curve OA', Metastable equilibrium:** The vapour pressure curve OA as liquid phase terminates at O, when the liquid is converted into solid. However it is possible to cool the liquid below the point O without solidification as shown by the curve OA'. This means that the liquid can be cooled far below the freezing point or 'supercooled' without solidification. This liquid would be in an unstable condition. On slightest disturbance or by addition of a small amount of seed crystal, the entire liquid solidifies rapidly. Thus OA' represents a metastable equilibrium.

- (c) **Triple Point:** The three lines on the phase diagram intersect at a common point called the **Triple Point**. Triple point shows the condition under which all the three phases (solid, liquid, vapour) can coexist in equilibrium. Thus triple point can be represented as,



$$F = C - P + 2 = 1 - 3 + 2 = 0$$

which predicts that system has no degrees of freedom. At triple point both pressure and temperature are fixed. If we try to change temperature or pressure the three phases would no longer co exist and one of the three phases will disappear.

2.4 ONE COMPONENT SYSTEM

The complexity of one-component systems depend on the number of solid phases that can exist in the system. The simplest case is that in which only single solid phase occurs. Water system (at lower pressure) and carbon dioxide system are the examples of the simplest one component system. For a one component system the phase rule equation can be written as:

$$\begin{aligned} F &= C - P + 2 \\ &= 1 - P + 2 \\ &= 3 - P \end{aligned}$$

There may arise three cases.

Case I. When only one phase is present

$$\begin{aligned} F &= 3 - P \\ &= 3 - 1 \\ &= 2 \quad (\text{Bivariant system}) \end{aligned}$$

It can be completely defined by specifying the two variables temperature and pressure or both temperature and pressure can be varied independently. Therefore a single phase is represented by an area on P – T graph.

Case II. When two phases are present in equilibrium, the degree of freedom will be:

$$\begin{aligned} F &= 3 - P \\ &= 3 - 2 \\ &= 1 \quad (\text{Monovariant or Univariant system}) \end{aligned}$$

This means that the pressure can not be altered independently if the temperature is changed. In such a case pressure is fixed automatically for a given temperature. A two phase system is depicted by a line on a P – T graph.

Case III. When three phases are present in equilibrium, the degree of freedom will be:

$$\begin{aligned} F &= 3 - P \\ &= 3 - 3 \\ &= 0 \quad (\text{Nonvariant or invariant system}) \end{aligned}$$

This special condition can only be attained at a define temperature and pressure. A three phase system is depicted by a point on the P – T graph. At this point the three phases are in equilibrium and this point is termed as the Triple Point.

THE WATER System

Under normal conditions the system water is a three phase and one component system. The three phases are liquid, ice and vapour. All these are represented by one chemical entity (H_2O), hence it is one component system.

On the basis of experimental data obtained for the water system a plot of relationships between various phases namely solid ice, liquid water and water vapour under different conditions of temperature and pressure are shown in the Fig. 2.2.

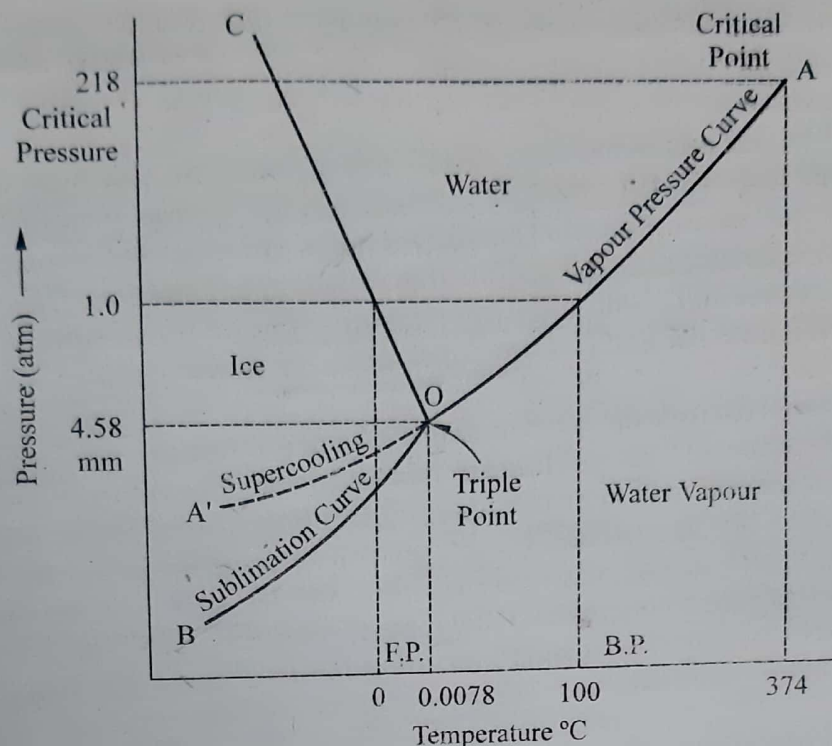


Fig. 2.2 The phase diagram of the system water

The three forms of water (ice, water and vapour) constitute the following equilibria

(i) **Single phase equilibria** *Area Phase*

- (a) Solid (ice) — represented by area BOC.
- (b) Liquid (water) — represented by area AOC.
- (c) Gas (vapour) — represented by area AOB.

(ii) **Two phase equilibria** *Curves*

- (a) Solid (ice) $\rightleftharpoons$ Liquid (water) — represented by curve OC.
- (b) Liquid (water) $\rightleftharpoons$ Gas (water vapour) — represented by curve OA.
- (c) Solid (ice) $\rightleftharpoons$ Gas (water vapour) — represented by curve OB.

(iii) **Three phase equilibria**

Solid (ice) $\rightleftharpoons$ Liquid (water) $\rightleftharpoons$ Gas (vapour) — represented by triple point O.

The phase diagram consists of:

- (a) Three areas, BOC, AOC, and AOB.
- (b) Three curves, OB, OA and OC.
- (c) One triple point, O.

The Areas

Area BOC represents the solid (ice) phase, which is thermodynamically most stable state under these conditions. Area AOC represents the liquid (water) phase in the system and similarly area AOB represents the gas (vapour) phase in the system. Thus these areas represent one phase equilibria.

In order to define the system completely at any point in the area, it is essential to specify both the temperature and pressure. Therefore areas have two degrees of freedom and are called bivariant systems. It can also be concluded by the phase rule equation.

$$F = C - P + 2 = 1 - 1 + 2 = 2 \text{ (Bivariant system)}$$

The Curves

The Curve OC (melting or fusion curve) represents the equilibrium between ice and water (two phase equilibria) and it is known as melting point curve. The very little slope of the curve shows that very high pressure is required to bring about significant changes in melting point of ice. The inclination of the curve OC towards the pressure axis indicates that melting point of ice slightly decreases by increasing pressure. This is the uncommon behaviour. The reason for this behaviour is the decrease in the volume, when ice melts into water. Thus ice transforms into denser water when the pressure is raised.

The Curve OA (Vapour pressure curve) represents the equilibrium between the two phases liquid water and vapour. Along this curve, water and vapour coexist in equilibrium. The curve shows the vapour pressure of liquid water at different temperatures. OA curve is also known as vapour pressure curve of water (l). This curve terminates at point A. This is the critical point having temperature 374°C and pressure 218 atm. At this critical point the liquid and vapour are indistinguishable from each other and thus only one phase is left. Critical point marks the highest temperature at which the liquid can exist. Critical point is a characteristic property of pure substance. When the vapour pressure is equal to 1 atmosphere, the corresponding temperature is the boiling point of water (100°C) as indicated by the phase diagram also.

Curve OB (sublimation curve) represents the equilibrium between ice and vapour and it is called as sublimation curve of ice. It shows the vapour pressure of solid ice at different temperatures. The two phases namely ice and vapour coexist in equilibrium along this curve. At the lower limit OB curve terminates at absolute (-273°C) where no vapour exists.

Along the curves OC, OA and OB, the number of phases present in equilibrium is two.

$$\begin{aligned} F &= C - P + 2 \\ &= 1 - 2 + 2 \\ &= 1 \end{aligned}$$

Hence, system is univariant along the curve.

Triple Point

The point O where all the three curves OC, OA and OB meet is known as triple point. At the triple point all the three phases of water system namely solid ice, liquid water, and gas vapour are in equilibrium. The equilibrium in three phases is attained at $.0076^{\circ}\text{C}$ temperature and 4.58 mm Hg pressure. Since, there are three phases and one component present in the system, the degree of freedom will be:

$$\begin{aligned} F &= C - P + 2 \\ F &= 1 - 3 + 2 \\ &= 0 \end{aligned}$$

Hence, the system is non variant. If either the temperature or the pressure or both are changed, the three phases would no longer coexist and at least one of them would disappear.

Metastable Curve OA'

The dotted curve OA' is the continuation of vaporisation curve OA. It is vapour pressure curve of supercooled water. As the water does not always freeze at 0°C , therefore if the vessel containing water and vapour is thoroughly clean and dust free, it is possible to super-cool water several degrees below its freezing point. On slight disturbance, the supercooled water at once changes to ice (solid). This kind of system is a metastable system.

Metastable State

It is the state in which the liquid below its freezing point is in the supercooled state which is not very much stable.

Supercooled liquid $\rightleftharpoons$ Vapour

On adding nucleus of ice (stable state), the system reverts to true stable system

Solid $\rightleftharpoons$ Vapour